Sahil Kumar

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Technical Skills

Programming Languages: C/C++, CUDA, Python, Java, JavaScript, SQL
ML/AI Frameworks: PyTorch, OpenCV, LangChain, LangGraph, LlamaIndex, Ollama
Vector Stores & Databases: FAISS, Chroma, Pinecone, PostgreSQL
Backend & App Frameworks: FastAPI, Flask-SocketIO, Flutter, Riverpod, Pydantic, Pytest
Embedded & Robotics: ESP-IDF, FreeRTOS, micro-ROS / ROS 2, Arduino, ESP32 / ESP32-S3
Tools & Platforms: Git, Docker, Linux, VS Code, GNU Make

Projects

NOIR: Autonomous Multimodal Robot Assistant | GitHub | *Python, PyTorch, VLM, Ollama, Flutter* 2025

- Built a mobile robot running three concurrent real-time loops (vision, voice, hardware control) in Python, with shared state behind a reentrant lock and deep-copied snapshots so no loop holds a mutable reference into another.
- Shipped a fully local voice pipeline (openWakeWord hotword, Faster-Whisper STT on CUDA, Gemma 4 via Ollama, Piper TTS) that also accepts microphone audio streamed from the phone app.
- Stabilized YOLOv8n person tracking with an EMA filter so bounding-box jitter never reaches the motors, and trained a MobileNetV2 classifier for line following.
- Hardened every hardware action: Pydantic-validated LLM output, allow-listed motor commands with bounded durations, and ~2,000 lines of unit, integration, e2e, and resilience tests.
- Wrote the ESP32 firmware in C on FreeRTOS (ESP-IDF v6) driving an L298N H-bridge over TCP, plus a Flutter app for live telemetry and manual control over WebSocket.

NyxCore: On-Device TinyML Watchdog for ROS 2 | GitHub | *C, ESP32-S3, micro-ROS, PyTorch* 2026

- Built an independent watchdog on a separate ESP32-S3 that monitors a robot's DDS streams and keeps reporting even when the main compute node stalls or crashes.
- Ran anomaly inference in pure C with no ML runtime: a Mahalanobis distance scorer (explicit 4x4 matrix multiply) and a sub-400-byte autoencoder trained in PyTorch, weights baked into firmware as static arrays.
- Scored each topic (`cmd_vel`, `scan`, `odom`) independently into a worst-case health level, with a 3-second silence watchdog catching dead publishers the statistical scores miss.
- Shipped it as a managed micro-ROS lifecycle node on FreeRTOS publishing `diagnostic_msgs` at 1 Hz, validated by a 9-script Python fault-injection harness (rate drops, jitter, message loss, silent topics).

Path Tracer: From-Scratch C++17 and CUDA Renderer | GitHub | *C++17, CUDA, Make* 2025

- Wrote a Monte Carlo path tracer twice with zero external dependencies: a single-file C++17 CPU reference (654 lines) and a CUDA port (586 lines) mapping one pixel to one GPU thread in 16x16 blocks.
- Ported the renderer to the GPU by turning the recursive bounce path into a bounded iterative loop accumulating attenuation, with device memory managed via `cudaMalloc/cudaMemcpy/cudaFree`.
- Implemented Lambertian, metal, and dielectric materials from the math (Snell refraction, Schlick Fresnel), jittered supersampling for anti-aliasing, and a thin-lens camera for real depth-of-field.
- Made renders reproducible with a `--seed` flag: Mersenne Twister on the CPU, inline per-thread XORShift on the GPU to avoid global-state contention.

Additional Projects

Kegel Trainer: Performance-Focused Flutter Wellness App | *Flutter, Dart, Riverpod (AI-assisted)* 2026

- Shipped a ~7,900-line Flutter app across 7 screens with zero bundled image assets; all art renders as inline SVG tinted from the active theme, scaling crisply at any size.
- Held frame rate stable on low-end phones by stripping blur and gradients from the render path, and cut the release APK from ~56 MB to 15–20 MB per ABI via R8 minification and ABI splitting.
- Built a bring-your-own-key AI coach over nine LLM providers that prompts from live app state, with user keys sent directly to the provider and never stored off-device.

Education

Deenbandhu Chhotu Ram University of Science and Technology Sonipat, Haryana
Bachelor of Technology in Electronics & Communication Engineering (IoT Specialization) *Nov 2022 – Present*

Experience & Achievements

Robotics Club Coordinator October 2023 – Present
TH!NKBOTS, DCRUST, Murthal *Murthal, Haryana*

- Coordinate club activities and mentor junior members on robotics fundamentals: Arduino programming, sensor integration, and project development.
- Developed club projects including a PID-controlled Line Follower bot and a Multi-Terrain Gripper bot.

E-Yantra Robotics Competition | *Team Leader* 2024, 2025

- **2024:** Led autonomous warehouse drone development using ROS2, OpenCV, and Gazebo simulation.
- **2025:** Developed self-balancing lunar scout robot with PID/LQR control systems and Fusion 360 design.

Smart India Hackathon 2024, 2025

- **2024:** Team member; built a water conservation cross-platform game using Godot Engine.
- **2025:** Team leader and event organizer, alongside competing as a member.

IEEE YESIST12 2026 | *Finalist* 2026

- Selected for the international final round in Indonesia with an AI-based cloud anomaly detection project.

Hackathon Finalist: Jewellery E-Commerce Platform 2025

- Built a full stack e-commerce website for a jewellery business and reached the finalist round.